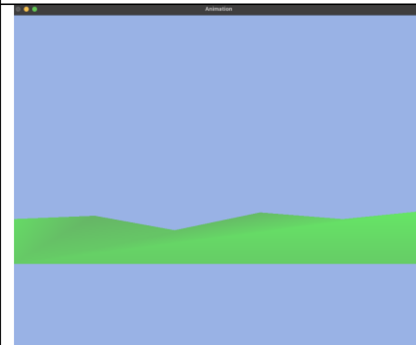
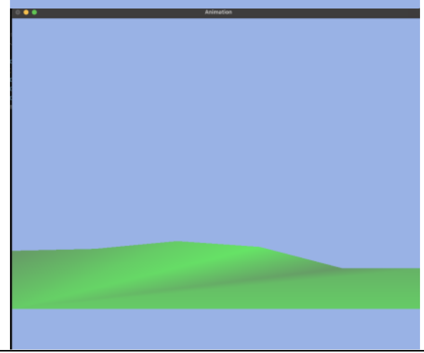
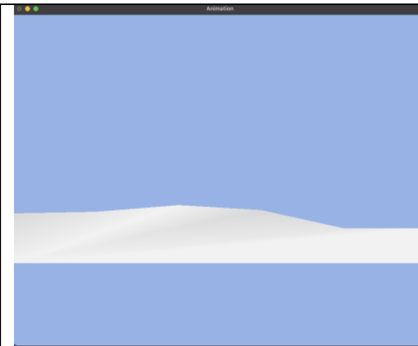
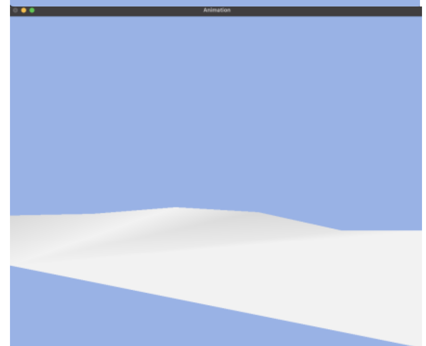
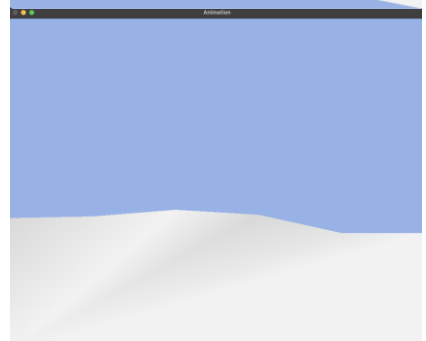
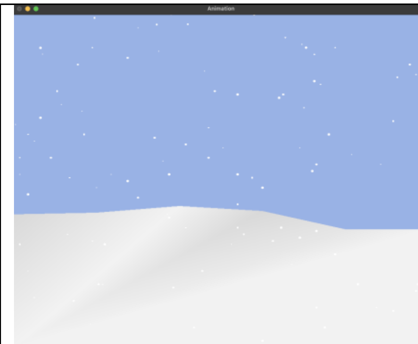
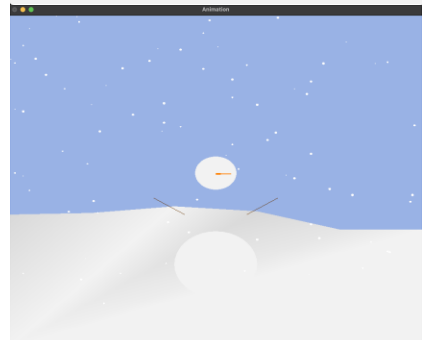
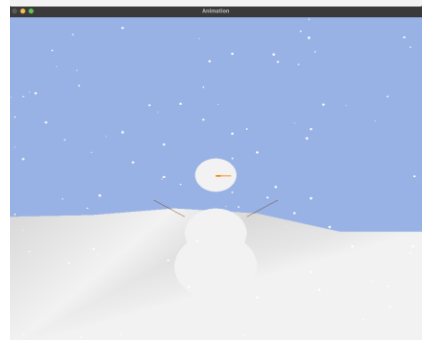
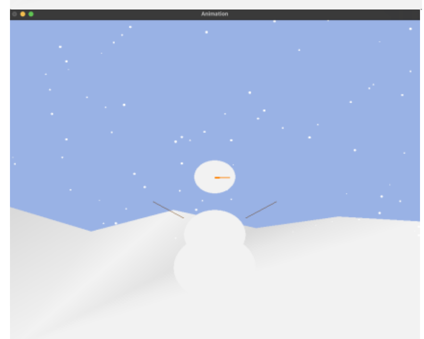
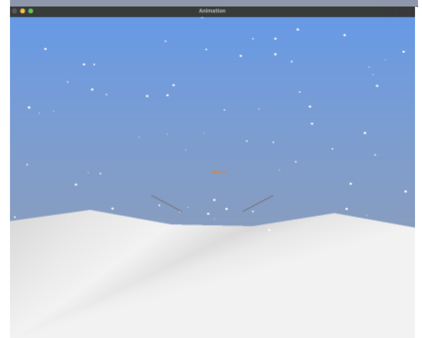
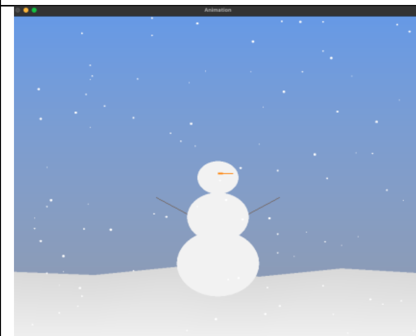
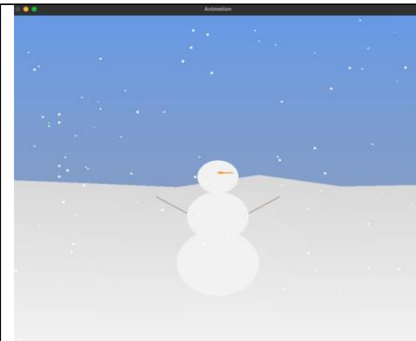
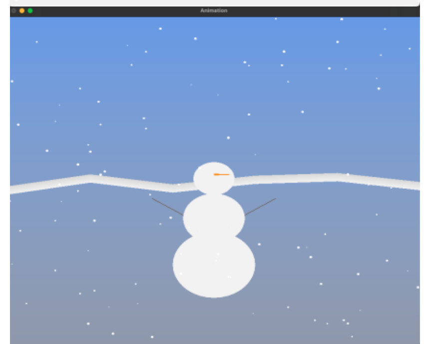
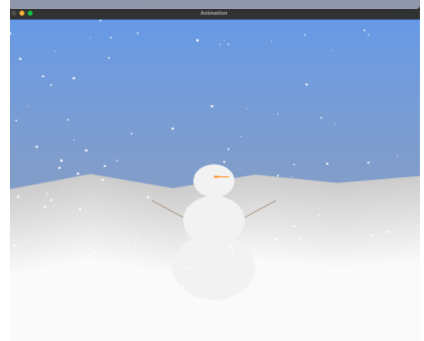
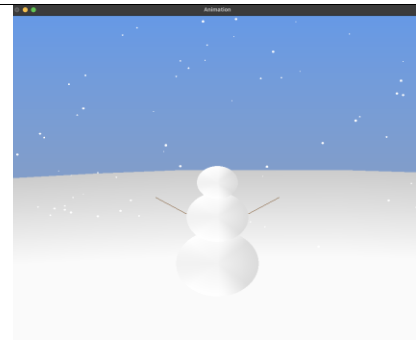
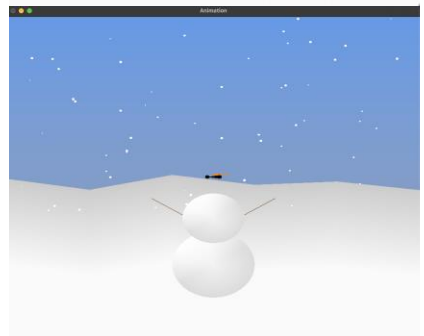
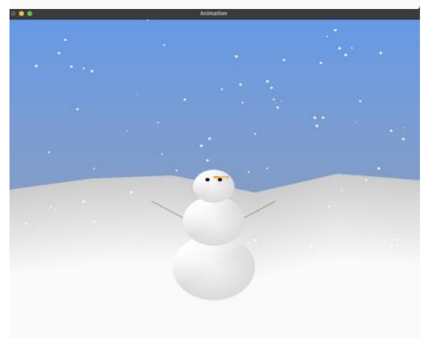
Duration	Date	To Do	Completed	Detail	Issues/Bugs
1 Hour	29-07-2025	- Figure out how to develop on Mac (code will later be converted for VS on windows)	Able to run program on VSC, will have to transfer everything over to VS2022 once I have access to a windows device.	n/a	n/a
2 Hours	01-08-2025	- Add a background color for the sky. - Add ground (color is chosen for better visual contrast due to the environment I'm working in being very bright).	Ground is made, sky is colored (will convert it to an object later).	 	The ground seems to be spinning at a rapid rate. Further diagnostics show that the hills I wanted to generate randomly were rapidly generating, giving the illusion of the ground spinning. I've changed it to a fixed integer rather than being random (due to spinning bug).

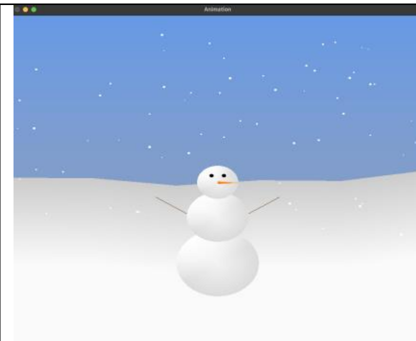
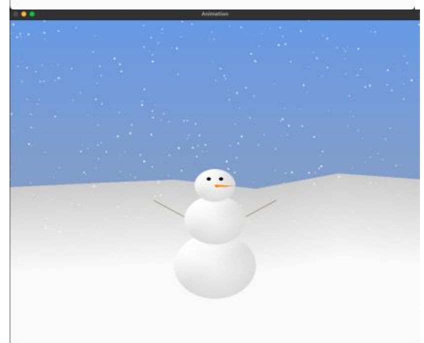
1 Hour	02-08-2025	- Connect the ground to the bottom of the frame (or connect the ground to the ground) - Present a more noticeable gradient for depth.	Ground is colored white Gradient is present, but can be improved (not suppose to use triangle, but playing around to learn, will change later).		Hilariously, every time I run "./snow_scene" it sets my computers key repeat delay to "long" in the settings. Not sure why, but I might investigate it later. After a few additions to my code, the ground is still "levitating" on the left corner. The shape and the gradient of the hill hasn't changed (which is good!), this indicates a minor issue in the code is stopping the left side extending down to the corner. Solution ended up being a minor oversight on my end. Simply changing 0.5f to 1.0f fixed the issue.
3 Hours	05-08-2025	- Add snow particles - Start drawing Mr Snow's body (2 circles)	Snow particles were successfully implemented with minor issues.		- Particles need to be re-sized (initially it is thick due to de-bugging and ease of knowing its presence).

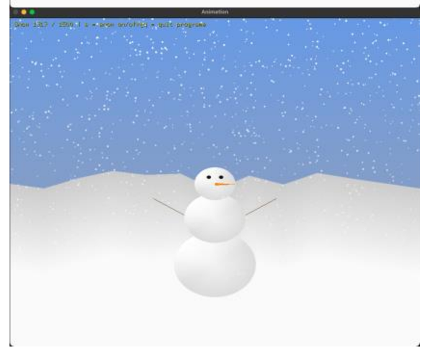
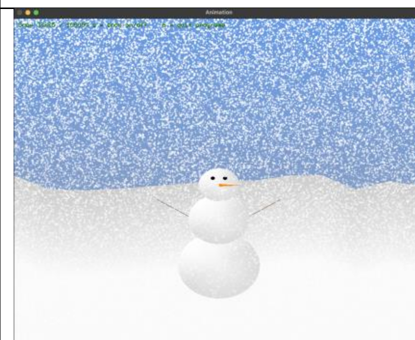
1 Hour	06-08-2025	- Fix layering issue (snow falling behind the “ground”) - Add key-bind to start/stop snow - Start drawing Mr Snow’s body (2 circles) - Ground to use random verticies	- Snow layering issue fixed - Mr Snow drawn (to a certain degree, still missing eyes). - Ground using random vertices	   	- Long key repeat delay still persists. - None of Mr Snow’s parts are appearing. Changing glVertex2f to 3f, making sure that Mr Snow isn’t hiding behind the ground/hill the issue still persists. - Mr Snow wasn’t appearing due to a typo “glVertext2f”, while the eyes and torso is missing, this is progress in the right direction. - Further debugging and I found out that I’ve put the same numbers for torso as I did for the legs, hence why it was appearing “absent”. Further2 tweaking is needed for Mr Snow to be better connected but the full picture is slowly coming together. - Ground is now able to generate random vertices each time with irregular polygons and not bug out like it had previously.

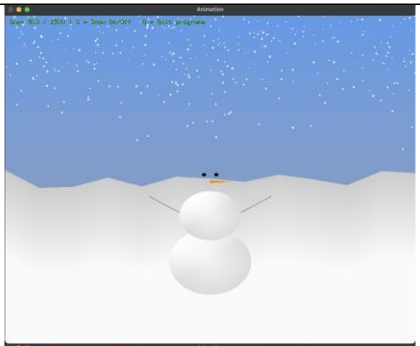
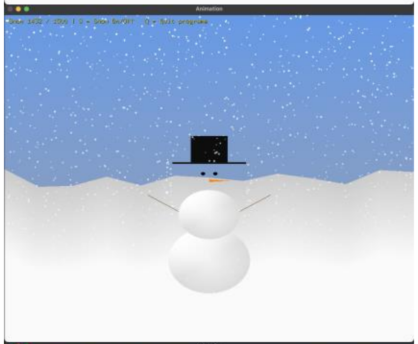
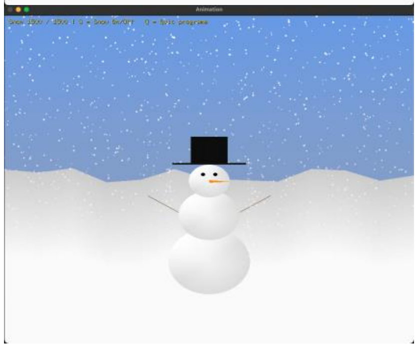
1 Hour	10-08-2025	- Draw sky as object and using vertex color and alpha blending (rather than a solid color as I did previously). - Complete Mr Snow and add gradient effect to him. - Remove triangle gradient from ground. - Make snow particles translucent.	- Sky drawn as object	 	- After sky is successfully drawn as object, the background and Mr Snows body + head have disappeared leaving just the carrot and arms. This may potentially be a layering issue as the snow particles are still present. - After putting the ground first in my display prototype, the issue is resolved confirming this was a layering issue on my end.
3 Hours	12-08-2025	- Add gradient to Mr Snow for a 3D effect. - Fix Mr Snows eyes (most likely a Z layering issue once again).	- Triangle gradient removed from ground - Mr Snow rendered in, with body heigh adjusted to the correct proportion. - Code for ground re-written to have adjustable gradient. - Mr Snows face finally place infront of his head, layering issue related to the arms caused the face to hide away.	 	- The ground was updated to generate the gradient via by-vertex. After a change I had made, the ground seems too short now to properly take affect of the depth. - Further efforts to extend it was successful, but the irregular hills generated each time the program ran is at such a small scale that its barely noticeable. - As usual, the issue is always a single typo. I had accidentally done multiplication over addition which completely shrunk my hills to nothing. - Particles are a little laggy, at this current point im still focused on sorting out the ground gradient and Mr Snow’s gradient + eyes, and will focus on particle efficiency later. I currently suspect there is an issue with my frame pacing, and might be better

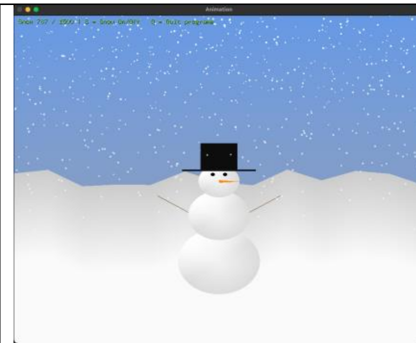
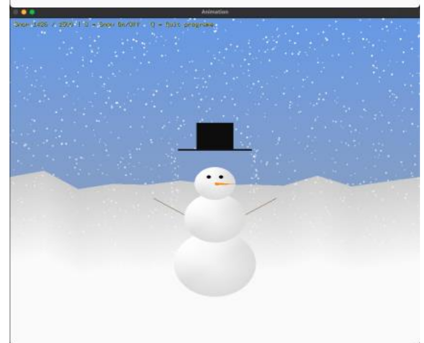
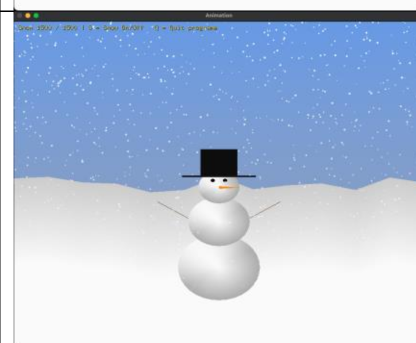
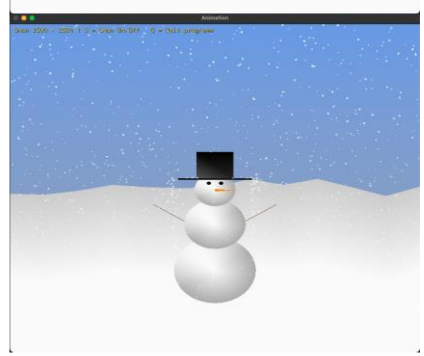
				  	off using a timer-based loop as I'm bouncing between 2 operating systems and need a guarantee that it will have similar performance on both. - I had an idea to tweak the gradient of my ground, but due to how I had written my code this wasn't possible and instead reduced the size of the ground. After re-writing my code to have the ground and the gradient as 2 separate things, I can fine tune the gradient density and spread with "const float gradD". - Few misinputs in my code (as usual) causing the gradient to be incorrect on Mr Snow. This is sorted out, but once again I'm facing a layering issue with the face (nose and eyes). - Commenting out Mr Snows head shows that the nose and eyes are present, but some layering issue is preventing it from appearing INFRONT of Mr Snows face.

6 hours	14-08-2025	- Sort out snow particle system.	Mr Snow now has a recognizable snowman face. Re-worked the snow particle system. (this included randomized spawns, spawning overtime, motions and effects such as swaying side to side, and gravity like speed). Particles drifting out of their left or right boundaries get repositioned to the opposite side (so ideally no particle is wasted out of the visible range). Opacity of snow flakes decrease as they approach the ground and slowly fade away (recycled). Added a function to enable and disable snowfall, need to add the keybind in animation.c. Also increased the particle count from 500 to 1500.	   	- A more advance line has been written for my particle system compared to what was written previously as a practice. The current issue however, is the snow particles spawning within a visible range making it unrealistic. - The previous issue stated about the snow lagging/slowing down in an unrealistic manner oddly resolved itself after trying to fix Mr Snows face layering issue. I cannot confirm what fixed the lag in the particles. - The particle count set to 500 takes too long to reach the ground and be recycled, leaving an awkward gap within the first 10 seconds where the snow stops. Increasing the particle count to 1500 fixes this issue and makes the scene look "richer". I understand this may not be the best solution, but this is a quick fix as I want to move on to the next question and potentially return to this later if I have enough time. - The bug faced on 2nd of August with my key repeat being set to "long" was due to me quitting the program via terminal. After setting ESC to quit the program, my key repeat is no longer set to long, yay! :D

	15-08-2025	- Add snow particle keybind (stop/start snow with "S" key). - Re-work the terrain(hill) generation - Maybe re-work the sky gradient?	Increased "NUM_GROUND_POINTS" from 6 to 13, to add more hills. (not sure if it looks better or worse, in the current screenshot it look decent. However other times it looks too jagged and sharp). Snow generation turned off with "S" and remaining snow gradually fall to the ground as they slowly de-spawn. Program can be closed with "Q" now instead of "ESC". ESC was only used during early stage of development, as it was easier for me due to muscle memory pressing ESC instead of Q. Display information ready and color coded (becomes more red the closer it gets to max particle number).	 		- Facing some issues with my HUD. Although I had some typos, even after fixing it my HUD still does not appear. I'll try changing it to a bright red color. Still nothing, even after trying to force the HUD be drawn over the sky, there is possibly a depth issue once again. - For over 4 hours no matter what i try, as long as its related to the snow count or max particle count, it will not update snow_scene if placed in display(). - Text is displayed, turns out I had to use think() instead of display(). However, now the screen is rapidly flashing. - After further investgation, i accidentally called glutSwapBuffer twice in display() which caused this issue. It's currently 8am and I've been programming all night, so it makes sense why I've been making such silly mistakes the whole night.
2 hours	19-08-2025	- 2 Additional features. (planning to add a top hat with collision physics, and wind controllable by arrow keys).		 		- Changed the particle from 1,500 to 100,000 out of curiosity. Screenshot was taken at 38,885 particles just for fun. - Top Hat has been implemented to Mr Snow, however my collision physics are incorrect. The snow just slides off the brim rather than bouncing off it. On top of that, when I press E to equip/unequip the hat, Mr Snow's face is despawned when the hat is equipped, and re-spawned when the hat is unequipped. - I've sorted out the layering issue between the face and the hat, but now gone back to the bug I previously had where the head is

				  		blocking the face, as ive had this issue before im certain its related to the face being depth occluded. - Issue still persistent, same issue as prior trying to get diagnostic text on screen. Using constant float for the Z causes the program to not actually update when running GCC even if i remove lines if code. Need to figure out what is breaking the code again. - As usual, a piece of code forgotten to be removed caused all this issue. After removing a const float in my zHat, everything worked perfectly!. This entire issue was caused by variable shadowing and my solution was to reorganize the Z values in a more consistent manner compared to what i had done before. - Added a new line to allow the hat have an ease of adjustment, but now the snow particles are hitting an invisible hitbox. Need to sort that out. - Made my hatAdjustment a global variable at the top of my mrsnow class, so that I can extern it in particle.c and use it in co-relation with my collision physics. This way, regardless of where the hat is placed, the collision will move with it (presented in a screenshot below, the hat is moved above Mrsnow and so has the collision successfully moved with it, only changing the value of “hatAdjustment” in mrsnow.c.

				 		
3 hours	20-08-2025	- Continue working on 2 additional features - Refine the gradients of sky and ground - Possibly refine Mr Snow's arms, nose and eyes?	Top hat has proper collision physics in relation to the snow particles. Some layer adjustments have been made where Mr Snow has snow falling behind him and Infront him (as per requirements of assignment). Top hat only has collision to snow particles falling infront of Mr Snow and ignores particles falling behind Mr Snow. Mr Snow's gradient has been refined once more to have a clearer 3D depth effect, making it easier to notice the snows falling behind him. Added diganostics mode for snow which shows which snow is front layer(green) and back layer(red). Improved sky gradient (once again), this time with 5 layers to make it appear very rich!. This had to be done due to my ground covering	 	- During the implementation of wind, and front layer collision with the Tophat, the layer of particles that suppose to fall behind mr snow is now falling infront of him.	

			Ground gradient has been improved, my previous ground lacked depth and looked a little too flat as I look back. I even asked a few friends to judge my gradient and most of them could hardly notice it, so it was added to a personal “to-do” list.